光電實驗二:雙視域之子相素源在柱狀透鏡陣列下之遠場分布

Light distribution in the dual-view system

【實驗儀器】

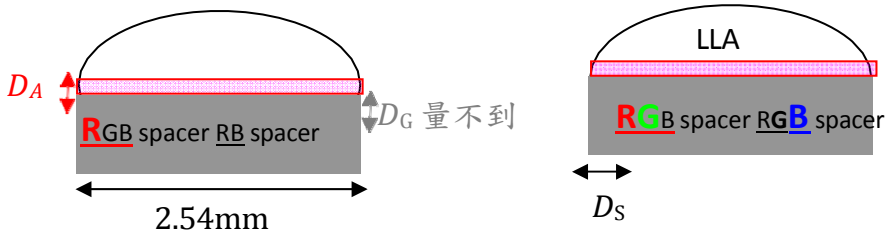
Mini LED 顯示器與控制器、影像輸入源、光譜照度計或亮度計 (OHSP350/350L)、Lenticular Lens Array (LLA 柱狀陣列透鏡，密度 10 LPI) 與量角器，手持式 CCD 影像觀測屏、防護鏡。

【實驗目的】

1. 觀察子像素源(sub-pixel)與 LLA 之空間相對位置，造成遠場角度、能量分布與半高寬等現象。
2. 參考範例程式於 page 3&4 完成交織圖像之建立。
   (**個人加分題**，需要寫類似程式對左右眼影片做分割)
3. 可參考以下影片與附錄資料
   https://youtu.be/fLz6gnuurPk
4. 了解雙視域實驗呈現靜態/動態裸眼 3D 圖像之機制。

【實驗步驟】

1. 以游標卡尺或 CCD 測量LLA 柱狀鏡的各項幾何參數。
   可假定 LLA 頂部為一圓弧。
2. 以自建軟體編碼製，基於 1 吋寬 x 2 英吋高之各色原圖，建構以下交織穿插之 two-view 底圖(a)(b)各一圖。
   (a) 紅/綠、紅/藍、綠/藍 (三原色光中取 2 色)
   (b) 黃/藍、紫/綠、靛/紅 (由a 中已選之 2 色混光，再取三原色中另一色。例如在 a 中選了紅/綠，則 b 這邊就選黃/藍。)



圖(一)：LLA 於Mini-LED 之示意圖

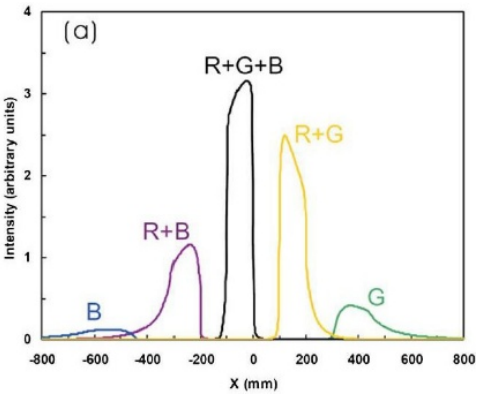
3. 由電腦分別輸出底圖(a)(b)投影到 Mini-LED 螢幕。每組 2 人，使用照度計 350 或者亮度計 350L (裝有雷射測距儀與 2°FOV 視角鏡頭)，並設法改變 LLA 與螢幕之間的距離(D_A [mm])作為實驗變數。儘量對齊，讓圖(一)中的$D_S \cong 0$。
4. 各組討論後，對螢幕與照度/亮度計取適當視距 L [mm]，測量各色光在不繞射階數出現峰值的角度，與這些峰值的角度半高寬。在前述峰值角度處，擷取光譜並輸出 excel 檔。計算各色光峰值之光譜積分面積(將照度/亮度對角度積分即為照度/亮度分布圖底下的面積，即可視為光強)。記錄各實驗條件下，各階繞射色光出現之角度、半高寬度與積分後得到的光強度。

   注意#1：L 指的是照度/亮度計與螢幕間的垂直距離，
   照度/亮度計的位置是在左右平移，但是指向 LLA 的中心。

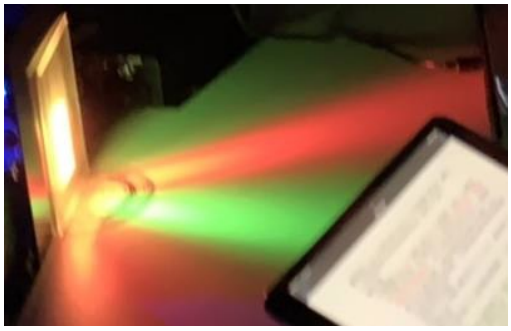
   注意#2：光譜儀各波長通道對應之 CCD 積分(收光)時間可由程式調整，請注意對各光譜數據比較時，要基於相同之積分時間。

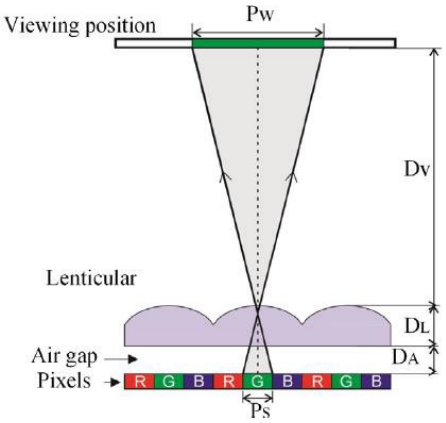
   注意#3：光譜儀 350 與 350L 有不同之收光空間角解析度。分析數據時，請注意並討論兩種光譜儀對於遠場光譜分析之差異性。

   提示：如右圖所示，同一組 RGB 在亮不同光點的時候會有不同的峰值分布。例如(a)中選了紅/綠，則會得到一些紅的峰值以及綠的峰值，接下來做 (b)黃/藍，會得到一些黃的峰值(會等於紅+綠嗎?)和藍的峰值。



5. 實驗觀察範例如下。注意各色光之不同階數(order)之空間能量分布、各色光出現角度與寬度。





Ref : “A spatio-temporal multiplexing multi-view display using a lenticular lens and a beam steering screen, “Optics Communications 420, 168–173 (2018)
注意：Mini-LED display 子像素分布與以上論文範例不同，同學可用手持式 CCD 觀察 MiniLED display 子像素分布。

範例程式一(matlab) 以紅綠二色為底本

編碼:程鴻昌助教 r12941068@ntu.edu.tw

```
%% Generating Multiview Image
colorred= [255,0,0]; % 舉例 宣告(RGB 色彩)紅色色座標
colorblue = [0,0,255]; % 舉例 宣告(RGB 色彩)藍色色座標
img = zeros(1080,1920,3); %創立 RGB 三通道 rows：columns=1080：1920 且數值為 0 的矩陣。
ppi=20; % pixels per inch(本次實驗 mini-LED 螢幕的像素密度)
lpi=10; % lens per inch(本次實驗 LLA 的像素密度)
k = 0;
row=1:100; % row range
for column =30:90% column range
   for channal =1:3 %channal=1 是紅色、channal=2 是綠色、channal=3 是藍色
      val = mod(column,ppi/lpi); % val = mod(i,2) dual-view slicing;
      switch val
         case 0 %將 mod 取餘數為 0 的情形改成紅色填入矩陣中
            img(row,column,channal)=colorred(1,channal);
         case 1 %將 mod 取餘數為 1 的情形改成藍色填入矩陣中
            img(row,column,channal) =colorblue(1,channal);
         otherwise %將 mod 取餘數不為 0 或 1 的情形
            k=+1;
      end
   end
end
img1=img(:,:,1); %確認在紅色通道中，column=30:90 之範圍中，符合奇數為 0；偶數為 255。
imshow(img); %matlab 即時顯示視窗
imwrite(img,'greenandredtest.png') %另存圖片檔，本次實驗需要。
```

範例程式二(matlab) 以紅藍二色為底本

編碼:程鴻昌助教 r12941068@ntu.edu.tw

```
clc;clear
%------2024/11/10-------
% ------定義圖像大小-------
rows = 1080; % 行數(螢幕縱向)
cols = 1920; % 列數(螢幕橫向)
rows_start=1;
rows_end=100;
cols_start=30;
cols_end=90;
% ------創建一個空的 RGB 數組------
maskData = zeros(rows, cols, 3); % 初始化為 0
% -----定義顏色並轉換到 [0,1] 範圍------
colorwhite=[255,255,255]/255;%白色
colorblack=[0,0,0]/255;%黑色
%------填充顏色(製作 mask)-------
for c = cols_start:cols_end %取列 30 到 90 這個範圍
   if mod(c,2) == 1 % 奇數列,mod 取餘數
      maskData(rows_start:rows_end,c,:) = repmat(reshape(colorblack, [1,1,3]),[rows_end-
rows_start+1,1,1]);
      %reshape(A,[m,n,x]) 將矩陣 A 重新依照 m 行 n 列排列，x 是三個顏色通道
      %repmat(A,[m,n,x])將 A 在行重複 m 遍；列重複 n 遍；顏色通道重複 x 遍
   else % 偶數列
      maskData(rows_start:rows_end,c,:) = repmat(reshape(colorwhite, [1,1,3]),[rows_end-
rows_start+1,1,1]);
   end
end
mask1=maskData;
mask2=maskData;
mask2(:,cols_start+1:cols_end+1,:)=mask2(:,cols_start:cols_end,:);
mask2(:,cols_start,:)=0;%確保原位置為 0
%----imread two picture------
[X, map,~] = imread('red.png');%uint8 單色會以灰階值形式呈現
red = ind2rgb(X,map); %uint8 轉 RGB uint8 三通道
[X1, map1,~] = imread('blue.png');
blue = ind2rgb(X1,map1);
targetSize = [rows_end-rows_start+1,cols_end-cols_start+1];%目標尺寸
redResized = im2double(imresize(red, targetSize, 'bicubic'));% 使用雙三次插值法
blueResized = im2double(imresize(blue, targetSize, 'bicubic'));
Image1=zeros(rows,cols,3);
Image2=zeros(rows,cols,3);
Image1(rows_start:rows_end,cols_start:cols_end,:) =redResized;
Image2(rows_start:rows_end,cols_start:cols_end,:) =blueResized;
%------切割及交織圖片------
Image1=Image1.*mask1;
Image2=Image2.*mask2;
Image_part=Image1+Image2;%疊圖
imshow(Image_part)
imwrite(Image_part,'redandblue.png')
```

實驗範例 三 (python) 動態讀取檔案並左右分割兩眼雙視域並合併

助教 張端麟 b11202053@ntu.edu.tw

【實驗報告與作業】

助教：

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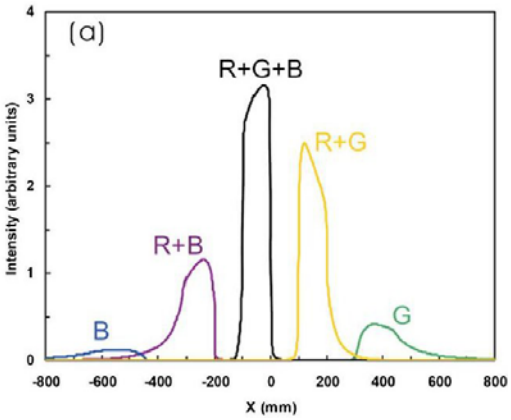
I. 實驗報告(每組交一份，總共 50%)

(a) 請記錄你們所量得之 LLA 各項幾何係數，以及你們所取的視距 L [mm]。 [5%]

(b) 以照度/亮度計與 LLA 中心的水平距離為 X 軸(所以照度計/亮度計與 LLA 中心位置的實際距離是 $\sqrt{L^2+x^2}$ ，量測到的光譜照度/亮度為 Y 軸作圖。請將各色光的作圖同時疊在一張圖上，page2 第 4 點提示中(a)狀況作一張圖，(b)狀況作一張圖。 [10%]
範例如下圖。請標明每個峰值是來自哪個顏色(如左邊的圖)以及來自左/右哪個 pixel。



(c) 討論實驗步驟 4 中，在 LLA 與螢幕距離D_A ≅ 0 [mm]的情況下，比較變化子像素點亮位置，所測量之光譜與角度位置的差異。例如 R 與 G 分別於不同像素源點亮(圖(一)的左圖情況)與在相同像素源位置點亮(圖(一)的右圖情況)，R 的峰 值、G 的峰值與 R+G 的峰值之間有什麼關係？ [15%]

(d) 討論實驗步驟 4 中，改變 LLA 與螢幕之間的距離D_A [mm]後，各色光不同階數之繞射角度與半高寬角會出現哪些變化趨勢？兩個 view 會分得更清楚或者更模糊？ [15%]

繳交格式：報告以小組為單位交於 COOL 上，命名為 Week1_Team#.pdf
例如第 01 組就是 Week1_Team01.pdf。

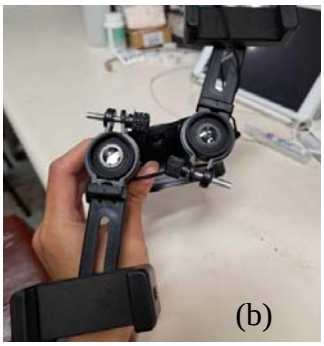
II. Part A 個人作業(50%) 與 Part B 加分題(100%)
報告要檢附程式碼並與助教約實機看影像

Part A (50%): 助教 程鴻昌 r12941068@ntu.edu.tw

參考範例一二，改寫為自己程式碼，利用 60 LPI 之 LLA 配合個人電腦/平版，製作多視域圖片 (N>2)詳 NTU COOL

Part B [個人加分題 100%] 助教: 張端麟 b11202053@ntu.edu.tw
參考範例程式三並改寫為自己程式碼+ 60 LPI LLA 製作裸眼 3D 動畫(至少 5 秒)
(1): 使用工業雙鏡頭攝影機拍攝 20m 以外之活動影像，作為模擬左右眼所觀察影像。
(2): 利用望遠鏡視角差，以同款之兩 CCD 聚焦於望遠鏡目鏡處，再用電腦幾乎同時驅動 CCD 對望遠鏡聚焦拍攝影片。後製時左右兩眼取相同幀數，製作裸眼 3D 動畫。
(3)比較步驟 1,2 後製裸眼 3D 影片效果，解釋你認為立體感差異原因與硬體上改進方案

(關鍵字 3D Lenticular animation，注意片長至少 5 秒長)
提示: https://www.youtube.com/watch?app=desktop&v=6MvYDOFiqLE
https://www.youtube.com/watch?v=YSgSVyvmFLQ



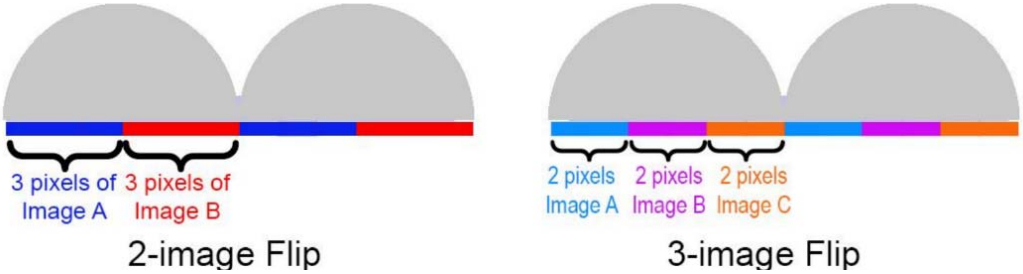
背景:宏碁 SpatialLabs 產品線，透過筆電螢幕上層的液晶柱狀透視鏡，投射 2 張些微不同的圖像分別到用戶左右眼形成裸視 3D 立體效果。
華碩 Spatial Vision，透過雙凸透鏡和先進的眼球追蹤攝影，為兩眼構築不同畫面，進而產生 3D 影像。

2a. 圖像分割重組前，
利用透明刻度片與手持式 CCD (跟助教借)，確認 LLA 之 LPI 與所使用電腦螢幕設定之 dpi 。
https://www.youtube.com/watch?v=ktgbtoCD58I
2b. interlacing software
圖像分割與重組，使用個人作業程式。
欲美化介面可參考以下格式
https://vuethru.com/lentlensblanks_superflip.html

3.一個 LLA 透鏡底下要放幾個像素? https://www.vicgi.com/lenticular-printing-guide-for-printers.html

Let's assume the true LPI (of LLA) is 50.1. Now, the crucial question arises: How many pixels should be placed under each lens or lenticule? While there isn't a fixed answer, a general rule of thumb is to ensure that when you multiply this number by the LPI, the result should be 300 or higher. In simpler terms, the smallest number of pixels should be the largest even integer that, when multiplied by the LPI, equals 300.

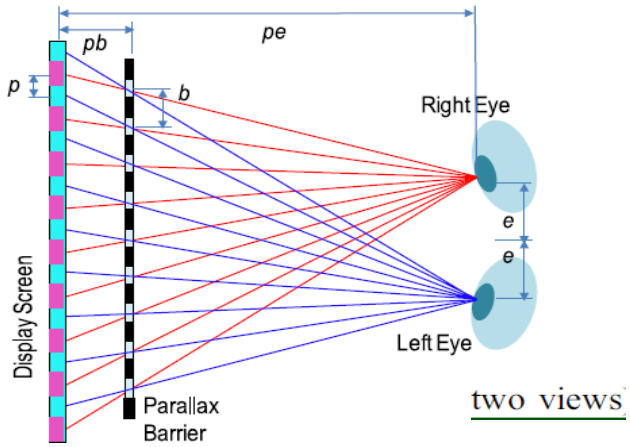
In the case of our 50.1 LPI, the calculation is 300 / 50.1 = 5.988. Rounding up to the next even number, we get 6.



【附錄與參考資料】

Ref 1: “Three-dimensional display technology,”
Advances in Optics and Photonics 5, 456–535 (2013) doi:10.1364/AOP.5.000456

A. Parallax-barrier (視差障蔽):



微雙視域範例
螢幕與障蔽間隙:空氣

若眼睛沿障壁方向移動，雙視域影像是否重現? 會出現 image flip 嗎?
需特定角度才能觀測嗎?影像強度變化如何?

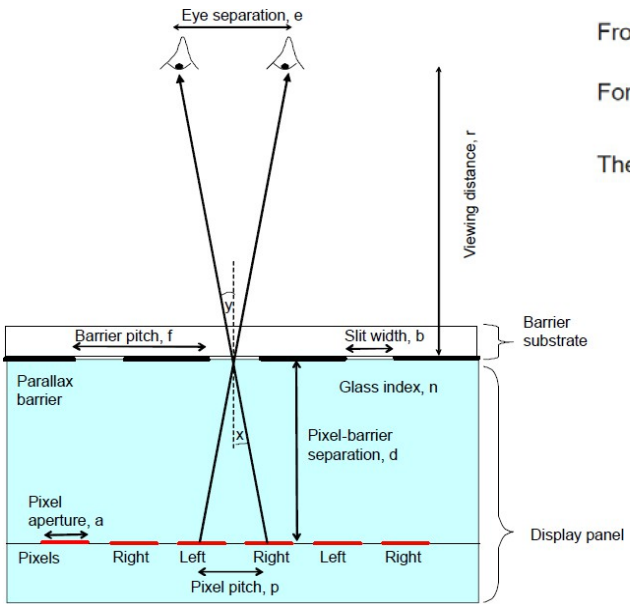
若把parallax-barrier 換成 LLA?

Usually, the stand-off distance pe and the pixel pitch p of the screen are pre-determined by hardware and viewing conditions. “e” is half of the eye separation distance. Based on similar triangle geometry, the appropriate barrier pitch b and the mask distance pb can be computed with

$$pb = \frac{(p * pe)}{(e + p)}, \qquad b = \frac{2 * p(pe - pb)}{pe}.$$

Drawbacks in the parallax-barrier-based displays, see p. 476

https://en.wikipedia.org/wiki/Parallax_barrier



From Snell's law: $n \sin x = \sin y$

For small angles: $\sin y \approx \frac{e}{2r}$ and $\sin x \approx \frac{p}{2d}$.

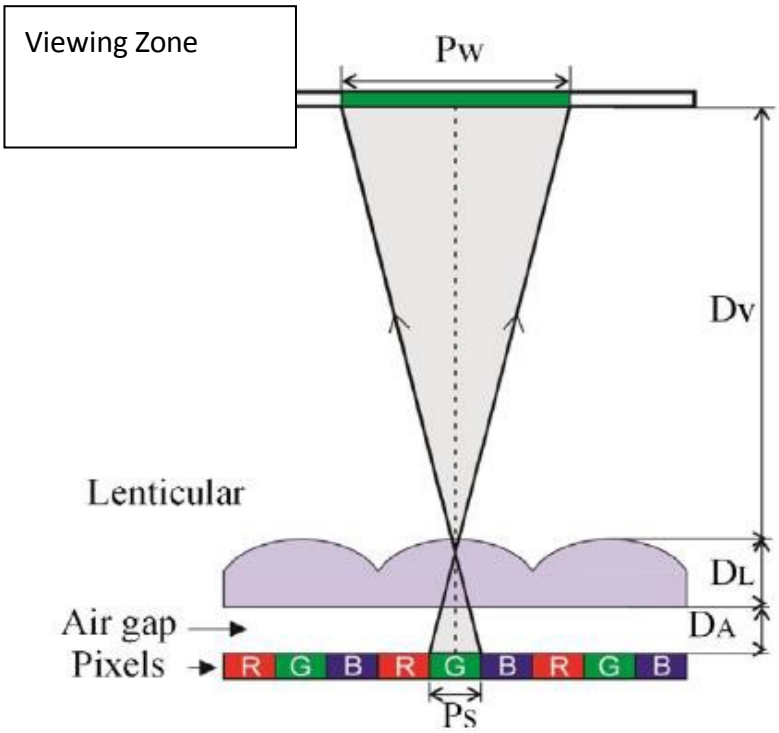
Therefore: $d = \frac{rnp}{e}$.

螢幕與障蔽間隙: 介質 n

e/p = r/(d/n)

Ref 2: “A spatio-temporal multiplexing multi-view display using a lenticular lens and a beam steering screen, “ Optics Communications 420, 168–173 (2018)

LLA 是一個各向漏光元件抑或它是特定光程導向元件? LLA 週期一定要是 pixel 像素尺度的整數倍嗎? 優化設計?



Another example: the air-gap lenticular sheet.
To reduce the weight of the lenticular sheet in the proposed prototype, it is designed to be mounted with an air gap as shown

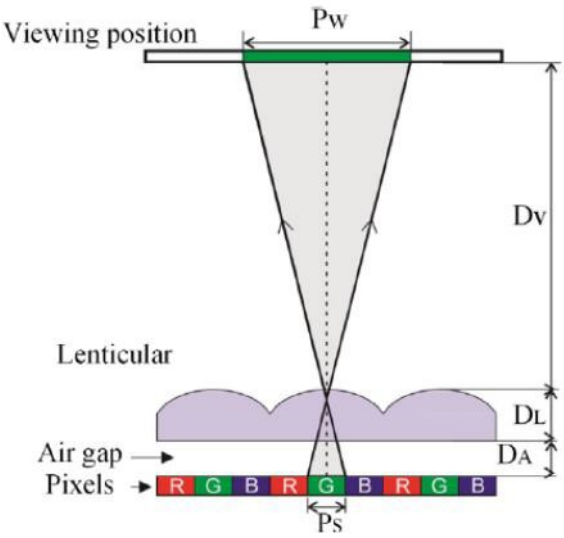
逆向工程思考:
In a conventional lenticular based autostereoscopic display, a lenticular sheet is directly attached to a display screen, and each cylindrical lens cell covers multiple sub-pixel columns. Since sub-pixels on the display panel are located at the focal plane of the lenticular sheet, they are magnified by the lens to form separate viewing zones at a certain distance, referred as the optimum viewing distance. The number of the viewing zones equals to the number of sub-pixels covered by each lens.

The viewing zone formation based on the proposed lenticular sheet is illustrated in Fig. 4. The optical magnification factor M is defined as the ratio between the width of the viewing zone P_W and the sub-pixel pitch P_S, which is given as follows:

$$M = \frac{P_W}{P_S}. \tag{1}$$

In order to get a magnified image at the optimum viewing position, the lens width should be slightly smaller than the sub-pixel pitch, and it can be calculated as follows:

$$P_L = \frac{k \times P_S \times M}{(M+1)}$$ where k denotes the desired number of viewing zones.



Parameter	Value
Number of views k	9
Pixel size	276.75 μm × 276.75 μm (RGB)
Lenticular screen size	531.36 mm (H) × 298.89 mm (V)
Sub-pixel pitch P_s	92.25 μm
Refractive index N_L	1.56
Viewing zone width P_W	20 mm
Optimum viewing distance D_V	1200 mm
Thickness of substrate D_D	2 mm
Radius of lens R	3109.6 μm
Height of lens D_L	27.5 μm
Pitch of lens P_L	826.6 μm
Air gap D_A	4.2 mm

Based on a geometrical approximation at the optimum viewing distance shown in Fig. 4, the air gap D_A between the lenticular sheet and the display screen can be calculated by the following expression.

$$D_A = \frac{D_V}{M} - \frac{D_L}{N_L} \tag{2}$$

where D_V is the optimum viewing distance while D_L and N_L correspond to the thickness and refractive index of the lenticular screen respectively.

The effective distance F of the lens cell is given as follows:

$$F = D_A + \frac{D_L}{N_L} + D_D \tag{3}$$

where D_D is the thickness of the glass substrate on the lenticular screen.

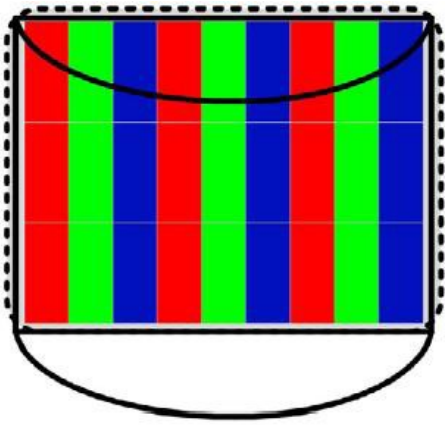
Thus, the radius R of the lenticular screen can be given as below:

$$R = F \times (N_L - 1)\,. \tag{4}$$

定義:觀測單位與螢幕之垂直距離為視距(viewing distance)
定義:水平方向之受光觀測區間為 viewing zone
人眼視角解析度極限(1/60) ~$\tan^{-1}$[錐細胞直徑/水晶體焦距]
~$\tan^{-1}$[(0.0045mm)/17mm] CIE standard observer: (2)FOV ~color-sensing cones are within a 2° arc of the fovea

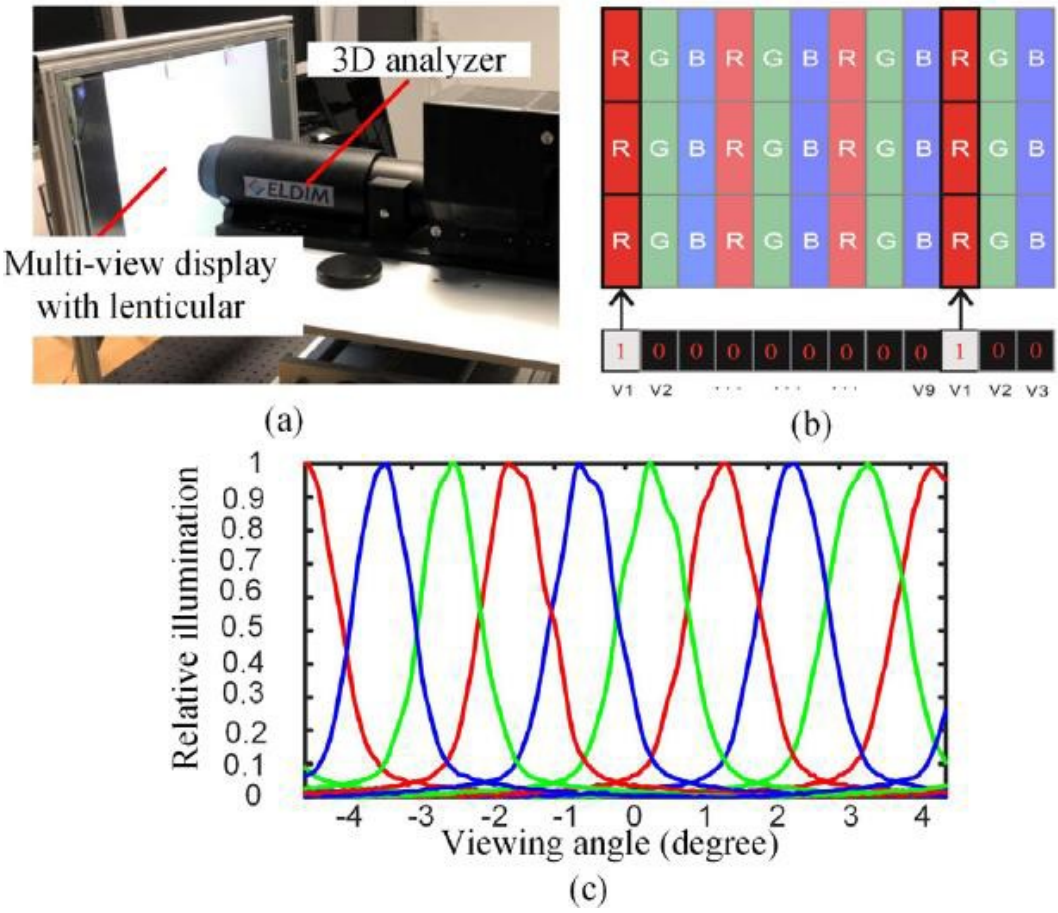
Optics Communications 420, 168–173 (2018) LLA 螢幕特徵測量

created in the horizontal space. However, in a given viewing position, only a monochrome image can be observed. In order to provide full color images, a temporal multiplexing technology is developed to elaborately shift images among the entire viewing field in sequence, and thus every viewer can continuously observe separate red, green and blue images.

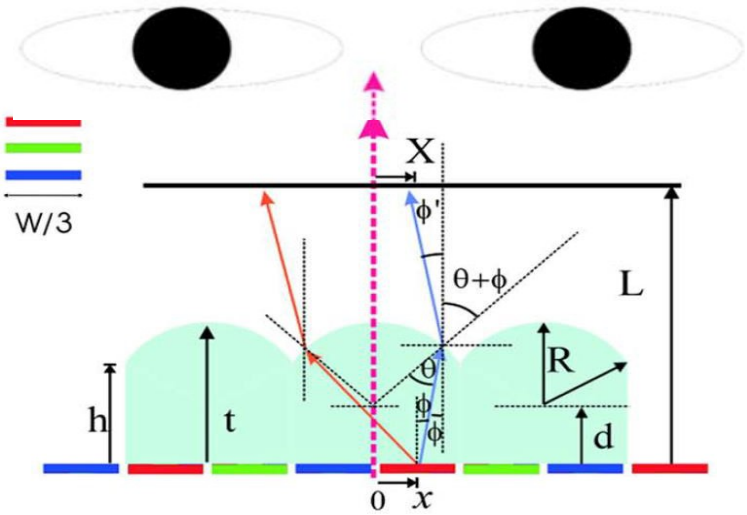


本例柱狀鏡下有 9 sub-pixels，水平方向重複出現 9 repeating viewing zones.

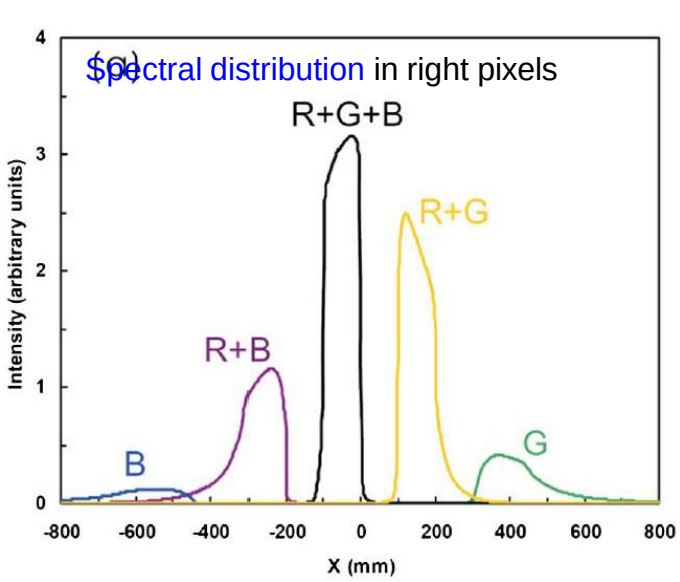
LLA 螢幕特徵測量

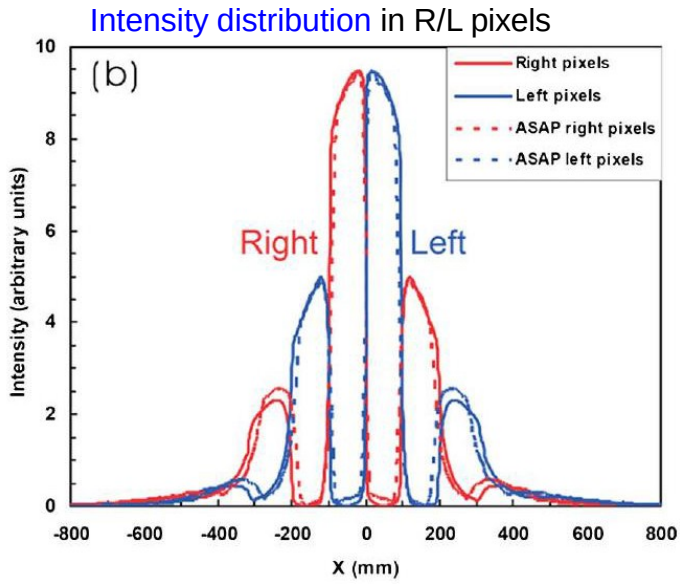


Ref.3 Y-C. Chang et al., “Efficient simulation of intensity profile of light through subpixel-matched lenticular lens array for two- and four-view auto-stereoscopic liquid-crystal display,” Appl. Opt. 52, A356 (2013); ibid 53, 5640 (2014)



duced. Figure 2(b) shows the total intensity profiles due to light from the right (in red) and left (in blue) pixels, and the dashed line is the simulation result obtained by using the commercial ray-tracing software ASAP. The parameters used are $n_g = 1.511$, $L = 350$ mm, $W = 0.27$ mm, $R = 0.18$ mm, and $h = 0.5$ mm. In the current simulation, we have assumed that the LLA has the same refractive index as the glass plate of the LCD, and the thickness of the



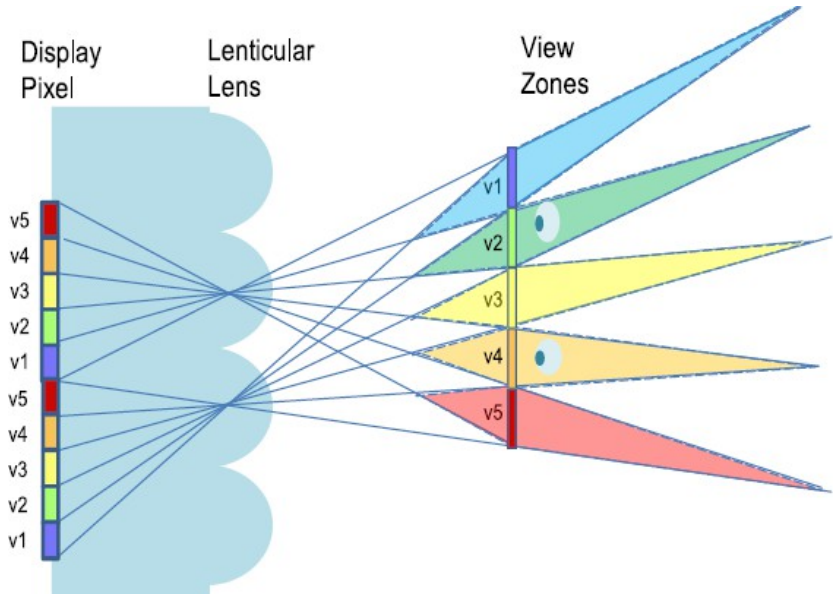


補充 Slant LLA

Ref: Advances in Optics and Photonics 5, 456–535 (2013)

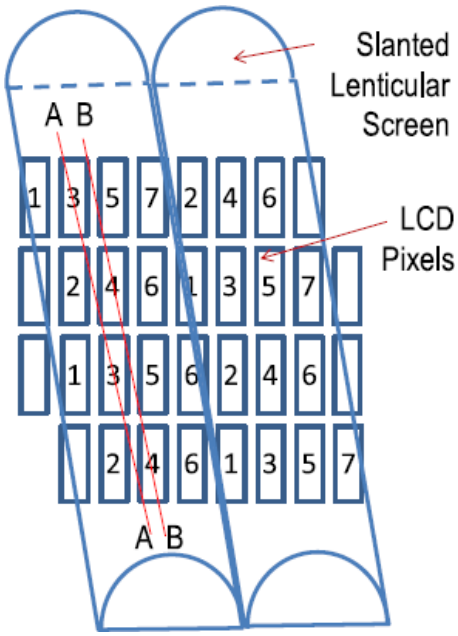
Lenticular Sheet type

Composed of plano-convex cylindrical lenses which are optically analogous to that of a parallax barrier screen. The optical efficiency is much higher than its parallax barrier counterpart



illustrates a HPO spatial multiplex display with five views using a lenticular lens sheet in front of a 2D LCD screen. The lenticular lenses are aligned with the vertical pixel columns on the 2D LCD screen. A pixel column is assigned to a single view for every five columns. If a viewer positions his or her eyes in the correct view zone, he can actually see stereo images from the spatial multiplex screen. When his head moves, motion parallax can also be experienced.

- **Limited resolution.** For a display with N views, the resolution of an individual view is essentially $1/N$ of the original display resolution.
- **Alignment.** Aligning a lenticular sheet with a screen requires significant effort.
- **Cross talk between views and image flips.** This may result in one eye seeing the image intended for the other eye, causing the human brain to perceive the stereo effect incorrectly.



Since it is obvious that the resolution of a 3D image is reduced in lenticular lens systems, a number of advanced techniques have been developed to compensate for it. One of them, developed by Berkel and co-workers [51,52] at Philips Research Labs, is a slanted lenticular system to distribute the loss of resolution into both the horizontal and vertical directions by slanting the structure of the lenticular lens or rearranging the color filter of the pixels.

Figure 27 shows the relationship between the pixels and the slanted lenticular sheet for a seven-view display. As the LCD is located in the focal plane of the lenticular sheet, the horizontal position on the LCD corresponds to the viewing angle. Therefore all points on the line A–A direct view 3 in a given direction, and all points on line B–B direct view 4 in another direction. The way in which the effect of flipping is reduced is evident by examining line A–A, where view 3 predominates, but with some contribution from view 2. Similarly, for the angle corresponding to line B–B, view 4 predominates with some contribution from view 3.

A number of 3D TV products based on the slanted lenticular screen on LCD techniques are currently available in the commercial market, including manufacturers such as Vizio, Stream TV Networks, Alioscopy, RealD, Philips, Sharp, Toshiba, and TLC. Some of them are working on 4 K UHD and even 8 K versions of 3D TVs.

51. C. van Berkel, D. W. Parker, and A. R. Franklin, "Multiview 3D LCD," Proc. SPIE **2653**, 32 (1996).
52. C. van Berkel and J. A. Clarke, "Characterization and optimization of 3D-LCD module design," Proc. SPIE **3012**, 179 (1997).